

Introduction to Statistical Methods - Mid-Semester 1 Mock 01 Solutions

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ISM Midsem 1 Mock 01 - Answer Key

Q1

A parameter is a numerical description of the full population; a statistic is calculated from a sample. One valid pair is population mean journey time μ for all monthly bus journeys and the 120-journey sample mean $\bar{x}$. Other correctly matched journey-time pairs, such as population variance with sample variance, are equally valid.

Q2

Mutually exclusive events satisfy $A \cap B = \emptyset$, hence $P(A \cap B) = 0$; they cannot co-occur. Independent events satisfy $P(A \cap B) = P(A)P(B)$; knowing one occurs does not change the other's probability. If both probabilities are positive, mutually exclusive events are dependent. A permutation counts ordered selections, $P(n, r) = n! / (n-r)!$; a combination counts unordered selections, $C(n, r) = n! / [r!(n-r)!]$. The deciding condition is whether exchanging selected positions creates a distinct result.

Q3

The joint count for Route R and delay is 32, so $P(R \cap D) = 32/200 = 0.16$. Total delayed is $32 + 18 = 50$, so $P(D) = 50/200 = 0.25$. Because $P(D) > 0$, $P(R|D) = P(R \cap D) / P(D) = 0.16 / 0.25 = 32/50 = 0.64$. Also $P(R) = 80/200 = 0.40$; since $0.16 \neq 0.40(0.25) = 0.10$, the events are not independent. Thus 16% of all shipments are Route R and delayed, 25% are delayed, and 64% of delayed shipments used Route R. Probabilities are unitless and lie in $[0, 1]$.

Q4

The modeling choice says that within Urgent, and separately within Routine, knowing one named feature does not change the other feature's conditional probability. It does not claim unconditional independence. Urgent score:

$$0.30(0.80)(0.60) = 0.144$$

Routine score:

$$0.70(0.25)(0.20) = 0.035$$

The evidence normalizer is $0.144 + 0.035 = 0.179 > 0$. Therefore $P(\text{Urgent}|x) = 0.144 / 0.179 \approx 0.8045$ and $P(\text{Routine}|x) = 0.035 / 0.179 \approx 0.1955$; choose Urgent. Inputs are valid probabilities and class priors sum to 1. The posterior is model-based: correlated features or poorly estimated likelihoods can make it poorly calibrated.

Q5

Normalization gives $k(1+2+3) = 6k = 1$, hence $k = 1/6$. The masses are $1/6, 2/6, 3/6$, all nonnegative and summing to 1 on support $\{0, 1, 2\}$.

$$E[X] = 0(1/6) + 1(2/6) + 2(3/6) = 8/6 = 4/3$$

$$E[X^2] = 0^2(1/6) + 1^2(2/6) + 2^2(3/6) = 14/6 = 7/3$$

$$\text{Var}(X) = 7/3 - (4/3)^2 = 21/9 - 16/9 = 5/9$$

Because X counts support incidents in an hour, its mean is $4/3$ incidents per hour and its variance is $5/9$ squared incidents for that one-hour count. Both moments exist because support is finite.

Q6

The story asks for successes among a fixed $n=8$ trials, each sensor has one of two outcomes, trials are independent, and every trial has constant $p=0.25$. These conditions and support $\{0,1,\dots,8\}$ justify choosing $X \sim \text{Binomial}(8,0.25)$.

$$\begin{aligned} P(X=2) &= C(8,2) (0.25)^2 (0.75)^6 \\ &= 28 (0.0625) (0.177978515625) \\ &\approx 0.31146 \end{aligned}$$

$E[X]=np=8(0.25)=2$ sensors. $\text{Var}(X)=np(1-p)=8(0.25)(0.75)=1.5$ sensors squared. Under the model, about 31.15% of repeated groups have exactly two failures; the long-run average is two failures per group.

Q7

Because σ is unknown and s is supplied, use the t interval:

$$\bar{x} \pm t^* s/\sqrt{n}$$

$SE=9/\sqrt{36}=9/6=1.5$ seconds. Margin $=2.030(1.5)=3.045$ seconds. The interval is $72.0 \pm 3.045 = (68.955, 75.045)$ seconds. Conditions are an independent random sample and a reasonable normal-population model for exact t inference. In repeated sampling, 95% of intervals built by this method would cover fixed μ when conditions hold.

Q8

Formula, substitution, and numerical calculation are not applicable because the claim lacks a complete stochastic model. Three days are exposures, not three binary trials, and incident counts can exceed three, so binomial support is unsuitable. If incidents followed a constant-rate Poisson process with independent increments and unchanged exposure, a three-day count could use Poisson with $\lambda=3 \times 5=15$ and support $\{0,1,2,\dots\}$. Without those assumptions, even that choice is not justified.

Q9

Sort the data. If n is odd, exclude the overall median and take the median of each remaining half; if n is even, split into equal halves and take each half's median. $IQR=Q_3-Q_1$; fences are $Q_1-1.5IQR$ and $Q_3+1.5IQR$. Fences are calculated cutoffs, whereas whiskers normally end at the most extreme observed non-outliers. Pearson's second coefficient $3(\bar{x}-\text{median})/s$ is undefined when $s=0$, and zero does not prove symmetry; inspect the graph.

Q10

With replacement, the composition resets, so identical draws can be independent under random mixing. Without replacement, the first draw changes the composition and usually changes the next probability. This between-draw dependence differs from mutual exclusivity within one draw: for example, "red" and "blue" on a single token cannot co-occur, regardless of what happens on a later draw.

Conceptual-bank checking paths

Use these only after attempting C01-C24. A complete response should follow each stated path.

- C01: Population = all lamps produced next month; sample = 80 tested lamps. Choose a lifetime parameter, such as population mean μ , and match it to the sample mean $\bar{x}$ from the 80.

- C02: Trace the outlier's effect: the high value pulls the mean and SD strongly; median and IQR depend on order positions and are more resistant. Therefore report median/IQR for robust center/spread, while noting the outlier rather than hiding it.
- C03: First check conditions. Approximate normality permits the empirical 68-95-99.7 approximation. Any distribution with finite positive variance permits Chebyshev's guaranteed lower bound $1-1/k^2$ within $k>1$ SDs.
- C04: For one die, outcome 4 corresponds to elementary event {4}; {2,4,6} is composite; $S=\{1,2,3,4,5,6\}$; $\emptyset$ contains no outcome.
- C05: Exclusivity gives $P(A \cap B)=0$; independence gives $P(A \cap B)=P(A)P(B)$. With both probabilities positive, the product is positive, so both equations cannot hold.
- C06: If three finalists receive gold, silver, bronze, order changes the result: use a permutation. A three-person committee ignores internal order: use a combination.
- C07: Conditioning restricts the reference group to B, so $P(A|B)=P(A \cap B)/P(B)$. Division requires $P(B)>0$.
- C08: A joint cell concerns both categories; a marginal sums a row or column into the full-total denominator; a conditional divides relevant joint cells by the fixed row/column total.
- C09: Verify the branches are mutually exclusive and exhaustive. Only then do their disjoint paths add to the whole event probability.
- C10: Prior $P(C)$ is belief before x; likelihood $P(x|C)$ models data given class; evidence $P(x)$ sums all class paths; posterior $P(C|x)$ is updated class probability.
- C11: State $P(x_1, \dots, x_p | C) = \prod P(x_i | C)$: features are independent after class is fixed. This does not assert unconditional feature independence.
- C12: Shared evidence may be omitted only to compare class scores/argmax. Normalize scores to report posteriors. Logs turn products into sums and reduce numerical underflow.
- C13: A PMF assigns discrete point masses summing to one; a PDF is nonnegative density integrating to one; a CDF accumulates probability. For continuous X, $P(X=x)=0$ and interval probability is area.
- C14: $E[X]$ is the weighted center and $\text{Var}(X)=E[X^2]-E[X]^2$ the expected squared spread. The relevant sums/integrals must exist; variance requires finite second moment.
- C15: Independence implies zero covariance when the needed moments exist. Zero covariance alone does not imply independence.
- C16: Bernoulli is one binary trial; binomial counts successes across fixed n trials. Check two outcomes, independence, constant p, and fixed n.
- C17: Binomial has support $0, \dots, n$, parameters n, p, mean np, variance $np(1-p)$, and fixed trials. Poisson has support $0, 1, \dots$, parameter/mean/variance λ , and counts events per exposure.
- C18: Normal and t support all reals and are symmetric, with t heavier-tailed. Chi-square and F are nonnegative and generally right-skewed. Degrees of freedom change their shapes.
- C19: SRS gives each size-n subset equal chance; stratified sampling draws within every stratum and can improve subgroup representation/precision. Unequal sampling fractions require population-aware weights.

- C20: Population distribution describes individuals; observed sample distribution is one realized dataset; sampling distribution describes $\bar{X}$ across repeated samples.
- C21: Under iid finite-variance sampling, the standardized sample mean approaches $N(0,1)$ as n grows. Required n depends on skew, tails, and dependence, so 30 is not universal.
- C22: Over repeated valid samples, about 95% of intervals from the procedure cover fixed μ . After one interval is realized, frequentist μ is fixed rather than having 95% probability of lying inside.
- C23: Translate target; state support/units/denominator/design; check assumptions; choose rule; calculate; verify bounds/support/units; interpret without reversing conditions or overstating confidence.
- C24: Probability is unitless; mean, SD, and SE use observation units; variance uses squared units. A value outside model support exposes a wrong model or calculation.

Numerical-bank checking paths

Use these after attempting N01-N24; every line shows the formula-to-substitution route and final check.

- N01: $\bar{x} = (4+6+7+9+14)/5 = 8$; median = 7. Squared deviations sum $16+4+1+1+36=58$, so population variance $58/5=11.6$ and SD $\sqrt{11.6} \approx 3.4059$ measurement units.
- N02: $0.30(84)+0.25(76)+0.45(91)=25.2+19+40.95=85.15$; weights sum to one, so the result is score points.
- N03: Median = 8; lower/upper half medians give $Q_1=3, Q_3=11$; five-number summary (2,3,8,11,20). IQR=8; fences $3-12=-9, 11+12=23$; no value is beyond them.
- N04: $P(A \cup B) = .55 + .40 - .18 = .77$; neither $= 1 - .77 = .23$. Both lie in $[0,1]$.
- N05: Multiply independent choices: $5 \times 4 \times 10 \times 10 \times 10 = 20,000$ badges; repetition makes each digit position have 10 choices.
- N06: Committee $C(10,3) = 10!/(3!7!) = 120$; medals $P(10,3) = 10!/7! = 10 \times 9 \times 8 = 720$. Only medals distinguish order.
- N07: Returned total $24+6=30$. Thus $P(\text{return}) = 30/150 = .20$, $P(\text{online}|\text{return}) = 24/150 = .16$, $P(\text{online}|\text{return}) = 24/30 = .80$, and $P(\text{return}|\text{online}) = 24/90 = 4/15 \approx .2667$; denominators are positive.
- N08: Without replacement, $P(\text{two gold}) = (3/8)(2/7) = 3/28$. Complement route: $P(\text{at least one gold}) = 1 - (5/8)(4/7) = 1 - 5/14 = 9/14$.
- N09: Total failure $= .4(.03) + .6(.08) = .012 + .048 = .060$; $P(B|\text{failure}) = .048/.060 = .80$ because failure probability is positive.
- N10: Defect evidence $= .65(.02) + .35(.06) = .013 + .021 = .034$; posterior $L2 = .021/.034 \approx .61765$.
- N11: Scores: $R = .55(.7)(.4) = .154$; $S = .45(.3)(.8) = .108$; total $.262$. Posteriors $.154/.262 \approx .58779$, $.108/.262 \approx .41221$; choose R.
- N12: With add-one smoothing, $(0+1)/(30+1 \times 8) = 1/38 \approx .02632$. Without smoothing, $0/30 = 0$, so one unseen-word factor can zero the whole product.
- N13: Normalize: $k(1+2+3+4) = 10k = 1$, so $k = .1$. $F(1) = .1 + .2 = .3$; $E[X] = 0(.1) + 1(.2) + 2(.3) + 3(.4) = 2$; $E[X^2] = .2 + 1.2 + 3.6 = 5$; variance $= 5 - 2^2 = 1$.
- N14: $1 = \int_{[0,2]} cx \, dx = c[x^2/2]$ evaluated from 0 to 2, so $2c = 1$ and $c = .5$. Interval probability is $.25(1.5^2 - .5^2) = .5$. Also $E[X] = \int_{[0,2]} .5x^2 \, dx = .5[x^3/3]$ evaluated from 0 to 2, giving $4/3$.

- N15: Marginals: $P_X = (.3, .7)$, $P_Y = (.4, .6)$. $P(X=1|Y=0) = .30/.40 = .75$; $E[X] = .7$, $E[Y] = .6$. Independence fails because $.10 \neq (.3)(.4) = .12$.
- N16: $P(X=2) = C(7,2)(.2)^2(.8)^5 = 21(.04)(.32768) = .2752512$; $P(X \geq 1) = 1 - .8^7 = .7902848$; mean $7(.2) = 1.4$; variance $7(.2)(.8) = 1.12$.
- N17: Exposure triples, so $\lambda = 3(1.2) = 3.6$. $P(X=0) = e^{-3.6} \approx .0273237$; $P(X \geq 1) = 1 - .0273237 = .9726763$; mean 3.6; SD $\sqrt{3.6} \approx 1.89737$ calls.
- N18: $z = (94-100)/4 = -1.5$ and $(106-100)/4 = 1.5$; the supplied central-normal probability therefore gives $P(94 \leq X \leq 106) = .8664$.
- N19: $E[X] = 30$ g; $\text{Var}(X) = 12^2/36 = 4$ g²; $SE = 12/\sqrt{36} = 2$ g, under independent finite-variance sampling.
- N20: $SE = 9/\sqrt{81} = 1$ minute; margin $= 1.96(1) = 1.96$; CI $44 \pm 1.96 = (42.04, 45.96)$ minutes. The valid procedure covers fixed μ in 95% of repetitions.
- N21: $SE = 4/\sqrt{16} = 1$ mg; margin $= 2.131(1) = 2.131$; CI $25 \pm 2.131 = (22.869, 27.131)$ mg. Exact small-sample t inference uses the stated independent normal-population conditions.
- N22: Let X count failures. Fixed eight independent binary equal-.1 trials give $X \sim \text{Binomial}(8, .1)$, support $0, \dots, 8$; $P(X=0) = C(8,0)(.1)^0(.9)^8 = .43046721$.
- N23: The interval is $50 \pm 12 = 50 \pm 2(6)$, so $k=2$. Chebyshev guarantees at least $1 - 1/2^2 = .75$; no shape information justifies the empirical rule.
- N24: $SE = 15/\sqrt{100} = 1.5$; margin $= 1.645(1.5) = 2.4675$; 90% CI $62 \pm 2.4675 = (59.5325, 64.4675)$ units, width 4.935. The 90% claim is repeated-procedure coverage.

Historical-style-bank checking paths

- H01: Inclusion-exclusion gives $P(A \cap B) = .48 + .37 - .68 = .17$. Since $P(B) = .37 > 0$, $P(A|B) = .17/.37 \approx .45946$.
- H02: Variance is $(\sqrt{1.92})^2 = 1.92$. From $np = 4.8$ and $np(1-p) = 1.92$, divide to get $1-p = 1.92/4.8 = .4$, hence $p = .6$ and $n = 4.8/.6 = 8$; support $0, \dots, 8$, with fixed independent binary constant- p trials.
- H03: $1 = c[(5-1) + (5-2) + (5-3) + (5-4)] = 10c$, so $c = .1$. Masses $.4, .3, .2, .1$ are valid; $F(2) = .4 + .3 = .7$; $E[X] = 1(.4) + 2(.3) + 3(.2) + 4(.1) = 2$.

Additional-bank answer explanations

These answers correspond only to C25-C26 and N25-N27 in the two answer-free drill banks; check them after attempting those items.

- N25: The number of sixes is $\text{Binomial}(20, 1/6)$ on $\{0, \dots, 20\}$ because there are 20 fixed, independent, binary, equal-probability trials.
- N26: A Poisson variable has variance equal to its mean, so $\text{Var}(X) = 4$; $P(X=0) = e^{-4} \approx 0.0183$.
- N27: $SE = \sigma/\sqrt{n} = 12/\sqrt{36} = 2$, in the observations' units.
- C25: Strong skewness or heavy tails can slow convergence of the standardized sample mean, so no universal sample-size cutoff applies.
- C26: In frequentist inference the parameter is fixed; under the assumptions, the random interval procedure covers it in 95% of repeated samples. A realized interval does not assign 95% probability to that fixed parameter.

ISM Midsem 1 Mock 01 - Point Rubric

Award points independently; accept equivalent methods. Arithmetic follow-through applies only to consistent, valid work.

Coverage totals. Sessions 1-6 carry 4 marks each and Sessions 7-8 carry 3 each. Forms total property 5, comparison 4, numerical 12, and scenario 9; numerical plus applied is 21 of 30. Difficulty is Easy 9, Standard 15, Stretch 6: $30 \times (0.30, 0.50, 0.20) = (9, 15, 6)$ exactly.

Q1

- Q1-P1 - 1 mark, Easy, Session 1, property or definition: Parameter describes a population; statistic is sample-computed.
- Q1-P2 - 1 mark, Easy, Session 1, property or definition: Gives any valid matching parameter/statistic pair for journey time, such as population mean μ with the 120-journey sample mean $\bar{x}$ (or population variance with sample variance).

Q2

- Q2-P1 - 1 mark, Easy, Session 2, comparison or distinction: Contrasts non-co-occurrence with the product test and notes positive-probability incompatibility.
- Q2-P2 - 1 mark, Standard, Session 2, comparison or distinction: Contrasts ordered permutations with unordered combinations and states the order condition.

Q3

- Q3-P1 - 1 mark, Easy, Session 3, numerical reasoning: Joint count, formula, and 0.16.
- Q3-P2 - 1 mark, Standard, Session 3, numerical reasoning: Delayed total, marginal formula, and 0.25.
- Q3-P3 - 1 mark, Standard, Session 3, numerical reasoning: Conditional denominator, substitution, and 0.64 in context.
- Q3-P4 - 1 mark, Stretch, Session 3, numerical reasoning: Independence calculation, probability conditions, and conclusion.

Q4

- Q4-P1 - 1 mark, Easy, Session 4, scenario analysis or interpretation: Applies conditional-on-class feature independence to both named features/classes and states valid prior/probability conditions.
- Q4-P2 - 1 mark, Standard, Session 4, numerical reasoning: Formula, substitutions, and scores 0.144, 0.035.
- Q4-P3 - 1 mark, Standard, Session 4, numerical reasoning: Normalizer 0.179, posteriors within ± 0.0005 , and selected class.
- Q4-P4 - 1 mark, Stretch, Session 4, scenario analysis or interpretation: Connects a conditional-independence or calibration caveat to the model-based result.

Q5

- Q5-P1 - 1 mark, Easy, Session 5, numerical reasoning: $k=1/6$; nonnegative masses sum to one.

- Q5-P2 - 1 mark, Standard, Session 5, numerical reasoning: Expectation $4/3$ with substitutions.
- Q5-P3 - 1 mark, Standard, Session 5, numerical reasoning: Second moment $7/3$ and variance $5/9$ with intermediate fractions.
- Q5-P4 - 1 mark, Stretch, Session 5, scenario analysis or interpretation: Interprets both moments with support, existence, and units.

Questions 1-5 account for 16 marks; Questions 6-10 account for 14 marks.

Q6

- Q6-P1 - 1 mark, Easy, Session 6, scenario analysis or interpretation: Selects $\text{Binomial}(8, 0.25)$ from the sensor story and justifies it with support and all trial conditions.
- Q6-P2 - 1 mark, Standard, Session 6, numerical reasoning: Binomial formula, substitution, intermediate arithmetic, and $P(X=2) \approx 0.31146$ within ± 0.00005 .
- Q6-P3 - 1 mark, Standard, Session 6, numerical reasoning: $E[X]=2$ and $\text{Var}(X)=1.5$ with units.
- Q6-P4 - 1 mark, Stretch, Session 6, scenario analysis or interpretation: Interprets the probability as a repeated-group proportion and the mean/variance with correct sensor-count units; checks probability bounds and nonnegative variance. Support and trial conditions are scored only in Q6-P1.

Q7

- Q7-P1 - 1 mark, Standard, Session 7, scenario analysis or interpretation: Chooses t from the story and explains the model/design conditions and 35 degrees of freedom.
- Q7-P2 - 1 mark, Standard, Session 7, numerical reasoning: $SE=1.5$, margin 3.045 , and interval $(68.955, 75.045)$ seconds with substitutions.
- Q7-P3 - 1 mark, Stretch, Session 7, scenario analysis or interpretation: Correct repeated-sampling interpretation without probability about fixed μ .

Q8

- Q8-P1 - 1 mark, Standard, Session 8, scenario analysis or interpretation: Reasoned N/A, rejects binomial by story/support, and identifies missing assumptions.
- Q8-P2 - 1 mark, Stretch, Session 8, scenario analysis or interpretation: Conditionally proposes Poisson, rescales to $\lambda=15$, states support, and limits the conclusion.

Q9

- Q9-P1 - 1 mark, Easy, Session 1, property or definition: Complete median-of-halves convention.
- Q9-P2 - 1 mark, Standard, Session 1, property or definition: Defines IQR/fences and distinguishes calculated fences from observed whiskers.
- Q9-P3 - 1 mark, Standard, Session 8, property or definition: States undefined-at-zero-SD and zero-not-proof-of-symmetry cautions.

Q10

Point	Award criterion
Q10-P1 - 1 mark, Easy, Session 2, comparison or distinction	Contrasts replacement and no-replacement probability behavior.
Q10-P2 - 1 mark, Standard, Session 2, comparison or distinction	Separates between-draw dependence from same-draw exclusivity and gives the requested valid example.